

TASK 2 – DEEP LEARNING IMAGE CLASSIFICATION PROJECT.

1.OBJECTIVE:

***To build an image classification model using (TENSORFLOW) and classify image from the fashion (MNIST)dataset.**

2.DATASET USED:

***Training image:60,000**

***Testing image:10,000**

3.PYTHON CODE:

```
# Import libraries
import tensorflow as tf
from tensorflow.keras.datasets import fashion_mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense
import matplotlib.pyplot as plt

# Load dataset
(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()

# Normalize data
x_train = x_train / 255.0
x_test = x_test / 255.0

# Build model
model = Sequential([
    Flatten(input_shape=(28,28)),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```
)

# Train model
history = model.fit(
    x_train,
    y_train,
    epochs=5,
    validation_data=(x_test, y_test)
)

# Evaluate model
loss, accuracy = model.evaluate(x_test, y_test)

print("Accuracy:", accuracy)

# Plot accuracy graph
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])

plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')

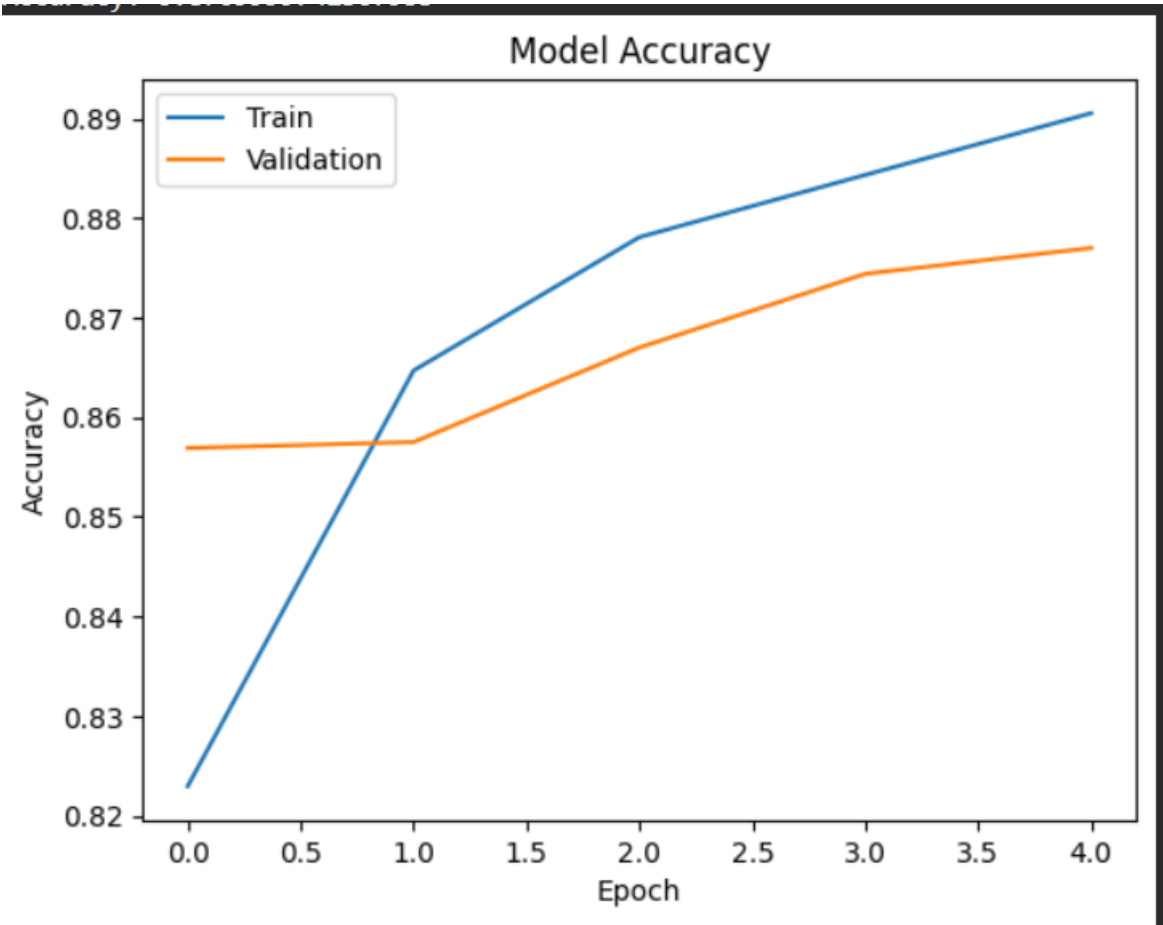
plt.legend(['Train', 'Validation'])

plt.show()
```

5.TRAINING AND OUTPUT SCREENSHOT:

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz
26421880/26421880 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz
5148/5148 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz
4422102/4422102 ————— 0s 0us/step
```

```
Epoch 1/5
1875/1875 ————— 12s 6ms/step - accuracy: 0.8229 - loss: 0.5011 - val_accuracy: 0.8569 - val_loss: 0.4123
Epoch 2/5
1875/1875 ————— 9s 5ms/step - accuracy: 0.8647 - loss: 0.3746 - val_accuracy: 0.8575 - val_loss: 0.3959
Epoch 3/5
1875/1875 ————— 9s 5ms/step - accuracy: 0.8781 - loss: 0.3379 - val_accuracy: 0.8670 - val_loss: 0.3714
Epoch 4/5
1875/1875 ————— 9s 5ms/step - accuracy: 0.8843 - loss: 0.3128 - val_accuracy: 0.8744 - val_loss: 0.3503
Epoch 5/5
1875/1875 ————— 8s 4ms/step - accuracy: 0.8905 - loss: 0.2957 - val_accuracy: 0.8770 - val_loss: 0.3521
313/313 ————— 1s 2ms/step - accuracy: 0.8770 - loss: 0.3521
```



6.RESULT:

Accuracy: 0.8769999742507935

7. CONCLUSION:

- *Successfully developed an image classification model using TensorFlow and Fashion MNIST dataset.
- *The model was trained and evaluated successfully and achieved 87.7% accuracy.